

Fast concept-based counterfactual explanations for image classification

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ABSTRACT

We propose **Fast Concept-based Counterfactual Explanations (FCCE)** for generating counterfactual explanations for CNN image classification models in real time. While counterfactual explanations are intuitive for humans, existing vision methods often face a trade-off between semantic interpretability and computational efficiency, limiting their widespread adoption in interactive decision support systems. FCCE addresses this challenge by operating in a semantic concept space and deriving a closed-form analytical counterfactual solution, with the remaining computational overhead dominated by standard feature extraction. To evaluate practical utility, we conducted a between-subject human study with 197 participants on a fine-grained model-failure identification task on the CUB dataset. Participants achieved 89.88% decision accuracy with FCCE, outperforming CVE (50.89%), ACE (68.47%), and ICE (82.95%), and reported higher perceived understanding, confidence, and satisfaction. Complementing the behavioral results, we assess fidelity and computational efficiency on CUB, MNIST, and ImageNet. Crucially, we demonstrate that FCCE counterfactuals remain highly faithful when mapped back to the original non-linear CNN, achieving a Relative Logit Gap below 0.3%, thereby providing strong empirical evidence for the fidelity of our counterfactual explanations. Finally, FCCE computes the analytical counterfactual solution in under 10^{-5} seconds on CPU, with an end-to-end local explanation overhead of approximately 10^{-2} seconds per instance. This is orders of magnitude faster than both optimization-based baselines and the representative source-conditioned diffusion-based generative counterfactual pipeline used as a latency reference. Overall, FCCE reconciles semantic interpretability with real-time performance and improves human decision-making about model reliability.

1. Introduction

The need for actionable interpretability. While deep learning models achieve high predictive performance, their internal decision-making strategies often diverge from human intuition [1]. To bridge this transparency gap, a wide range of interpretability techniques have been developed. Passive explanations (e.g., saliency maps) can highlight where a model focuses, but in high-stakes domains such as medical diagnosis and safety-critical systems [2,3], this is often insufficient. In such settings, users require actionable interpretability: explanations that not only describe a prediction, but also indicate how it could be changed [4]. Counterfactual explanations, which answer the question “why class P rather than Q ?”, are therefore widely regarded as a natural and effective paradigm [5–7]. However,

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in vision tasks, deploying counterfactual explanations remains challenging due to the tension between semantic interpretability and real-time responsiveness.

From pixel space to semantic concept space. To address these challenges, we argue for shifting counterfactual reasoning from the high-dimensional pixel space to a semantic concept space. Concepts (e.g., “beak color”) provide human-interpretable abstractions that bridge the semantic gap while simplifying the geometry of the decision boundary. In contrast, existing approaches struggle to balance semantic clarity and efficiency. Generative methods [8–10], including diffusion-based counterfactual explanation pipelines, operate in pixel space and can produce visually realistic counterfactuals. However, they often incur prohibitive inference latency due to iterative denoising, making real-time interaction difficult in practice. Sample-based and optimization-based methods [11–13] avoid direct generation, but rely on iterative search or large candidate pools, which hinders real-time interaction. Even existing concept-based approaches [14], while semantically grounded, often depend on segmentation quality or manual concept filtering, limiting scalability. In contrast, FCCE adopts unsupervised latent concepts learned directly from convolutional feature maps (following ICE-style factorization), avoiding explicit image segmentation and manual concept curation. These limitations highlight the need for an automated, concept-centric approach that supports both semantic interpretability and real-time use.

Proposed method and validation. To fill this gap, we propose Fast Concept-based Counterfactual Explanations (FCCE). FCCE is an unsupervised framework that projects inputs into a semantic concept space (via ICE [15]) and exploits the local linearity of class decision boundaries to derive a closed-form analytical counterfactual solution. Importantly, while FCCE relies on a locally linear approximation in concept space for efficiency, the resulting counterfactuals are ultimately evaluated by the original non-linear CNN. We therefore explicitly assess whether concept-space counterfactuals remain faithful after reconstruction in the original model, thereby addressing a central concern of surrogate-based counterfactual explanations.

We validate FCCE through a unified empirical study of human utility, fidelity, and computational efficiency. In particular, because empirical evidence on how concept-based counterfactual explanations support human decision-making in realistic tasks remains limited, we place special emphasis on human evaluation. In a large-scale human study ($N = 197$), participants using FCCE achieved significantly higher decision accuracy (89.88%) than those using CVE (50.89%), ACE (68.47%), or ICE (82.95%). Complementing these behavioral results, computational evaluation on CUB, MNIST, and ImageNet shows that FCCE counterfactuals remain faithful when mapped back to the original non-linear CNN, with a relative logit gap below 0.3%. Moreover, FCCE derives a closed-form analytical counterfactual in under 10^{-5} seconds on CPU, with negligible overhead beyond standard feature extraction, enabling real-time interaction without iterative optimization.

Our contributions are summarized as follows:

- We propose FCCE, an unsupervised concept-based framework that generates actionable counterfactual explanations via a closed-form analytical solution, avoiding iterative optimization and generative counterfactual sampling.
- We present a large-scale human study ($N = 197$), providing empirical evidence that FCCE significantly improves human decision accuracy (89.88%) and subjective understanding compared to state-of-the-art baselines.
- We provide a comprehensive computational evaluation, demonstrating that FCCE achieves real-time interactive performance, with the analytical counterfactual computation taking $< 10^{-5}$ seconds and an end-to-end local explanation overhead of approximately 10^{-2} seconds per instance, while maintaining high fidelity (relative logit gap $< 0.3\%$) to the original non-linear model.

2. Related work

2.1. Post-hoc explanations for vision models

Deep learning has achieved remarkable success in computer vision, exemplified by architectures such as VGG [16], ResNet [17], and Inception-V3 [18]. To understand these black-box models, a large body of post-hoc explanation methods has been proposed, primarily focusing on local feature attribution in the input space.

Pixel and region-based explanations. Early efforts to interpret CNNs focused on visualizing internal representations and salient regions [19]. Subsequent methods, such as saliency maps [20] and Class Activation Mapping (CAM) [21], highlight influential pixels or regions to reveal where the model attends. Other approaches, such as LIME [22], estimate feature importance over input-level segments. However, several studies have highlighted fundamental limitations of these methods [23]. While they can indicate where the model focuses, they often fail to explain what semantic factors drive the prediction. Individual pixels are rarely semantically meaningful [24,25], and raw image segments, although offering improved localization, may still not correspond to high-level abstractions that are intuitive to humans [26,27]. These limitations motivate the development of concept-based explanations.

2.2. Concept-based explanations

Concept-based explanations aim to bridge the semantic gap by representing model decisions in terms of human-understandable concepts rather than raw input features [28].

Supervised concepts. Early concept-based approaches, such as TCAV [26], Network Dissection [29], and IBD [30], rely on labeled data to bridge the semantic gap. Furthermore, Concept Bottleneck Models (CBM) [31] explicitly enforce an interpretable bottleneck layer aligned with ground-truth concepts. Despite varying definitions—ranging from linear directions to aligned basis components or bottleneck activations—these methods universally aim to project latent features into a semantic concept space. In this shared space, model decisions can be interpreted through concept scores. Their dependence on predefined concept vocabularies and extensive manual annotation limits scalability and applicability to unlabeled datasets [32].

Unsupervised concepts. To alleviate the labeling bottleneck, unsupervised concept discovery methods have been proposed. ACE [27] clusters image segments to automatically extract concepts, though its performance is sensitive to segmentation quality [33]. Alternatively, methods operating directly on feature map activations, such as CA [34] and ICE [15], apply matrix factorization to learn latent concepts. A common visualization strategy is to present prototypes—instances or regions with high concept activation—sometimes with highlighted spatial support [15]. Notably, ICE further approximates the deep model’s decision logic using a linear classifier over concept scores, enabling reasoning about class-level decision boundaries in the concept space. Despite these advantages, the reliance on Non-negative Matrix Factorization restricts the representation of inhibitory features. This specific constraint has been addressed by recent work on Archetypal SAE [35] using semi-NMF. FCCE builds upon this finding: instead of merely analyzing importance weights as in ICE, we exploit the linearity of this concept space to derive analytical solutions for counterfactual generation.

2.3. Counterfactual explanations

Counterfactual explanations describe the minimal change required to alter a model’s prediction. In the context of vision models, existing approaches can be broadly categorized by their search space and generation mechanism.

Optimization-based methods. Classical counterfactual methods formulate explanation generation as a constrained optimization problem in the input space, balancing validity, similarity, and sparsity [36–38]. While well-suited to tabular data, directly applying these techniques to high-dimensional image pixels often produces adversarial perturbations [39,40] that lack semantic interpretability [41]. Some papers aim for fast real-time methods [42], but they operate in the input space and do not provide concept-level abstractions.

Generative methods. Generative approaches employ models such as GANs [43] or diffusion models [44] to synthesize visually realistic counterfactuals [8,9]. Although capable of producing high-quality images, these methods incur substantial computational cost due to iterative generation procedures (e.g., denoising steps in diffusion models), often requiring seconds per instance. Moreover, pixel-level synthesis may introduce visually plausible edits whose relationship to the classifier’s decision logic is not always transparent, which can complicate their use in decision-support settings.

Latent space optimization and semantic constraints. An alternative line of work avoids direct pixel generation by optimizing in a latent space or applying semantic constraints. CVE [11] manipulates latent representations by replacing localized regions with samples from the target class (distractors). Other works incorporate semantic consistency constraints [12] or specialized regularization [13] to improve plausibility. However, these methods typically rely on iterative gradient descent or large candidate pools, which limits their suitability for real-time interaction.

Concept-based counterfactuals. Combining concept representations with counterfactual reasoning offers a promising direction for interpretable vision models. CACE [10] generates counterfactual explanations using causal concept effects, but relies on generative models (StyleGAN), inheriting their latency issues. CoCoX [14] generates counterfactual explanations using unsupervised concepts derived from image segmentation. Despite its semantic grounding, CoCoX depends on segmentation quality and often requires manual concept filtering to ensure interpretability. In addition, its search-based optimization procedure incurs non-trivial computational overhead.

Our distinction. FCCE operates in an unsupervised concept space and admits a closed-form analytical solution for counterfactual generation. In contrast to generative approaches with high inference latency and search-based methods that require iterative optimization, FCCE computes counterfactuals in constant time with respect to the number of concepts. As a result, FCCE is designed for real-time interactive explanation scenarios, where low latency and stability are first-order constraints.

3. The FCCE method

This section presents the theoretical framework of Fast Concept-based Counterfactual Explanations (FCCE). FCCE generates counterfactual explanations in a semantic concept space and admits closed-form solutions, enabling real-time interaction. We first introduce the concept-space linearization used by FCCE. We then define counterfactual objectives and derive analytical counterfactuals. Finally, we convert counterfactuals into sparse, human-understandable explanations and evaluate fidelity in the original CNN.

3.1. Background: latent space linearization

FCCE operates on the premise that complex CNN representations can be simplified into a semantic concept space. Whether this space is constructed via supervised directions (e.g., CBM [31]) or unsupervised factorization (e.g., ICE [15]), the subsequent decision logic can often be approximated linearly. This universality allows our analytical derivation to be applied across diverse concept definitions.

Given an input image x , a CNN classifier F is decomposed at a target layer l as

$$F(x) = H_l(G_l(x)).$$

Here G_l denotes the feature extractor and H_l denotes the classification head. Let $\mathbf{A} = G_l(x) \in \mathbb{R}^{W \times H \times C}$ be the feature maps at layer l . A concept learning method is applied to $\mathbf{A}$ to obtain a reduced concept feature map $\mathbf{S}$. Global average pooling is then used to obtain the concept score vector $s \in \mathbb{R}^{C'}$,

$$s = \text{GAP}(\mathbf{S}).$$

Under common CNN architectures (e.g., ResNet, Inception) that use GAP followed by a linear classification layer, the prediction function in concept space can be effectively approximated as

$$f(s) = w^\top s.$$

This linear decision assumption in concept space is widely adopted by concept-based explanation methods such as CBM [31], CA [34], and ICE [15]. We conduct counterfactual reasoning in this concept space rather than the pixel space for two reasons. First, concept scores s provide a low-dimensional, semantically meaningful representation, addressing the interpretability gap of pixel-based methods. Second, the linearity of $f(s)$ enables the derivation of closed-form analytical solutions, avoiding the computational cost of iterative non-convex optimization. Crucially, this analytical derivation is mathematically agnostic to the sign of concept scores, remaining valid for latent representations derived from semi-NMF or other methods that relax the non-negativity constraint. While this assumes a linear decision boundary in the concept space, our fidelity analysis (Section 3.6) confirms that this approximation holds well for standard CNNs.

3.2. Background: counterfactual objectives in concept space

Given an original instance s_0 with predicted class y and a target class y' , FCCE seeks a counterfactual instance s' in concept space. We summarize desired counterfactual properties using a multi-objective loss,

$$\mathcal{L}(s') = L_{\text{vali}} + \alpha L_{\text{simi}} + \beta L_{\text{spar}} + \gamma L_{\text{plau}}.$$

Validity L_{vali} enforces that s' crosses the decision boundary. For binary class comparison with weights w_o and w_t , defining $\Delta w = w_o - w_t$, we formulate the validity constraint as finding a root for $\Delta w^\top s' = 0$. Similarity L_{simi} penalizes deviation from the original instance, defined as the squared ℓ_2 norm $\|s' - s_0\|_2^2$. Plausibility L_{plau} constrains the counterfactual to lie near the target-class data manifold, approximated via distance to class centroids c_t . Sparsity L_{spar} aims to minimize the number of modified concepts to prevent cognitive overload, formally defined as the ℓ_0 norm $\|\Delta s\|_0$. Since directly optimizing the non-differentiable ℓ_0 term is computationally intractable in a closed-form solution, we relax this constraint during the projection phase and enforce sparsity via explicit feature selection (top- k filtering) in the explanation generation stage (Section 3.5). In the following derivation, we minimize the quadratic terms (L_{simi} and implicitly L_{plau}) subject to the validity constraint, handling sparsity as a post-processing step.

3.3. Analytical counterfactuals without plausibility

We first derive analytical counterfactuals considering validity and similarity only. We enforce validity by the decision boundary constraint and solve

$$\min_{s'} \|s' - s_0\|_2^2 \quad \text{s.t.} \quad \Delta w^\top s' = 0.$$

The constrained problem admits a closed-form solution via a Lagrangian formulation:

$$\mathcal{L}(s', \lambda) = \|s' - s_0\|_2^2 + \lambda \Delta w^\top s'.$$

Setting $\nabla_{s'} \mathcal{L} = 0$ yields the analytical projection

$$s' = s_0 - \frac{\Delta w^\top s_0}{\Delta w^\top \Delta w} \Delta w.$$

Geometrically, this corresponds to projecting s_0 onto the decision boundary along its normal direction. A toy example illustrating this geometric intuition is shown in Fig. 1. The computational complexity of this solution is $O(C')$, where C' is the number of concepts, yielding negligible overhead in practice.

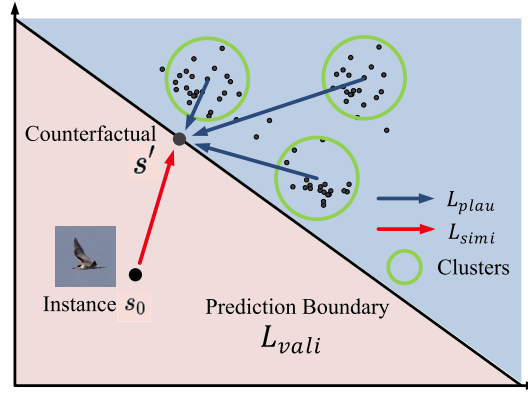


Fig. 1. A toy example for our method in concept space. With linear prediction space, the counterfactual instance s' is on the prediction boundary where $L_{vali} = 0$, represented by a direction (red arrow) starting from the given instance s_0 . We can adjust the red arrow (counterfactual direction) to satisfy sparsity or plausibility requirements. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

3.4. Plausibility as a design principle and its instantiation

While the above solution guarantees validity and minimal change, plausible counterfactuals should also remain close to the target-class data manifold. This desideratum can be formalized by the joint objective

$$\min_{s'} \|s' - s_0\|_2^2 + \gamma \|s' - c_t\|_2^2 \quad \text{s.t.} \quad \Delta w^\top s' = 0,$$

where c_t denotes the centroid of the target class in concept space. This objective characterizes the desired balance between similarity and plausibility. Rather than explicitly solving this constrained optimization, FCCE instantiates a closed-form and search-free construction that is consistent with the above objective. Importantly, we do not claim that the instantiated direction below is the exact optimizer of the above constrained objective. Rather, it is a closed-form surrogate construction motivated by the same similarity–plausibility trade-off, designed to preserve the analytical and search-free nature of FCCE. Under the centroid-based plausibility assumption, we define an instantiated conceptual change direction as

$$\Delta s_{inst} = w_t \odot c_t - w_o \odot s_0,$$

where $\odot$ denotes element-wise multiplication. This instantiation provides a stable heuristic direction that incorporates both target-class centroid information and classifier-dependent concept weighting, and is used throughout our experiments as a practical closed-form realization of the plausibility principle. When multiple centroids are used to approximate the target manifold (e.g., a Gaussian mixture model), we compute candidate directions $\Delta s_{inst}^{(j)}$ for each centroid and select the one with the lowest similarity–plausibility cost.

3.5. From counterfactuals to human-understandable explanations

The analytical counterfactual s' calculated in Section 3.3 specifies the exact projection onto the decision boundary. However, mathematically, this projection often results in small, non-zero changes across all concept dimensions (i.e., a dense change vector), which can be cognitively overwhelming. To satisfy the sparsity objective L_{spar} defined in Section 3.2 and ensure human interpretability, we apply a post-hoc filtering step to sparsify the explanation.

We use Δs to denote the conceptual change vector induced by a counterfactual. While this vector can be derived directly from the boundary projection ($\Delta s_{proj} = s' - s_0$), we prioritize plausibility in our framework. In our framework, the analytical projection derived in Section 3.3 provides the general counterfactual solution under the validity–similarity objective. The centroid-instantiated change introduced in Section 3.4 can be viewed as a closed-form instantiation that additionally reflects the plausibility design principle. In the experiments and explanation display, we use this instantiated form for constructing sparse, human-understandable concept rules.

The induced change in the decision function is given by

$$\Delta f = \Delta w^\top \Delta s.$$

To attribute this decision change to individual concepts, we define the concept-wise contribution vector

$$s'_{cont} = \frac{\Delta w \odot \Delta s}{|\Delta w^\top \Delta s|}.$$

This vector decomposes the decision change into normalized contributions of each concept. We then enforce sparsity by retaining only the indices i where $s'_{cont,i}$ is among the top- k largest contributions (or exceeds a cumulative threshold), masking others to zero. This step effectively optimizes the ℓ_0 constraint, converting the dense analytical solution into a compact, human-understandable concept rule that matches human attention limits.

3.6. Real-model fidelity analysis

Fidelity in FCCE involves two complementary aspects. First, at the surrogate level, one may ask whether the linear concept-space model derived from ICE provides a faithful approximation to the original non-linear CNN. This notion of concept-space fidelity has been discussed in prior ICE-based analysis and reflects the global agreement between the surrogate predictions and those of the original model over the data distribution. However, this surrogate-level fidelity alone is insufficient for our purpose.

FCCE is designed for counterfactual explanation, and therefore requires a second, more task-specific notion of fidelity: whether a specific counterfactual computed analytically in concept space remains faithful after being mapped back to the original CNN. Because ICE provides a reconstruction mapping from concept space back to the input representation, the concept-level counterfactual can be approximately reconstructed as an input instance and evaluated directly by the original CNN. Accordingly, we assess counterfactual fidelity on the real model using the original CNN. Let x' be an input instance aligned with the concept-level counterfactual. We evaluate x' using the original CNN F and denote the resulting logits for the original and target classes as z_o and z_t , respectively. Since a faithful counterfactual should lie close to the true decision boundary of the original model, these two logits should be nearly equal. We therefore define the Relative Logit Gap (*R-Gap*) as

$$R\text{-Gap} = \frac{|z_o - z_t|}{|z_o| + |z_t| + \epsilon} \times 100\%.$$

This normalized form is adopted because the absolute scale of logits may vary substantially across datasets, classes, and models, making raw logit differences less comparable across settings. A smaller *R-Gap* indicates that the reconstructed counterfactual lies closer to the true decision boundary of the original model, and therefore implies higher counterfactual fidelity between the concept-space solution and the original non-linear CNN.

3.7. Global explanations

Beyond instance-level explanations, FCCE naturally supports global explanations. By replacing the instance s_0 with class centroids and applying the same analytical construction, FCCE yields class-to-class counterfactual explanations in concept space. This provides global insights into how concept-level changes drive transitions between classes, and directly supports our global analyses in the experiments. In practice, these class-level counterfactuals form the basis of the global explanation interface used in the human study and the qualitative examples presented later in [Section 6](#).

4. Human behavioral study

To rigorously evaluate the utility of FCCE in real-world decision-making, we conducted a large-scale, between-subject human behavioral study. The study was designed to assess both the interpretability benefits of FCCE compared to state-of-the-art baselines and the effect of explanation information density on human performance. Specifically, we compared FCCE against three representative methods: CVE (non-concept, counterfactual) [11], ACE (segment-based concept, factual) [27], and ICE (latent concept, factual) [15]. While our primary focus is on evaluating FCCE as a global explainer, we additionally conducted an explanation information ablation experiment to analyze how varying the amount and type of explanation information (e.g., local explanations and textual feature labels) influences human decision accuracy and subjective perception.

4.1. Hypotheses

We formulate three hypotheses to guide the design and evaluation of our human behavioral study, focusing on both objective decision performance and subjective user perception.

H1: Interpretability. Participants using FCCE will more accurately identify whether a model prediction is correct or incorrect compared to those using CVE, ACE, or ICE. This hypothesis tests whether concept-based counterfactual explanations enable users to better understand the model's decision logic and reliably assess its failure cases.

H2: Information density. Providing additional explanation information, such as local (instance-level) explanations or textual concept labels, will improve decision accuracy compared to presenting global explanations alone. This hypothesis evaluates whether increasing the amount of explanatory information helps users make better decisions, or whether excessive detail may hinder performance.

H3: Subjective perception. Participants using FCCE will report higher subjective satisfaction, perceived understanding, and decision confidence compared to baseline methods. This hypothesis examines whether the objective benefits of FCCE are reflected in users' subjective experience with the explanations.

Rationale. An effective explanation should help users identify the reliable operating regions of a model, namely when its internal reasoning is likely to hold and when it is prone to failure. As illustrated in [Fig. 2](#), understanding which visual concepts the model relies on (e.g., leg-related features) allows users to detect systematic failure modes, such as cases where these features are absent or occluded.

Global Explanation:

Class [Caspian_Tern] is different from Class [Common_Tern] with around 31.90% difference. The absence/presence of the following 3 features make up 91.44% of this 31.90% difference, support the instances in Class [Caspian_Tern].	
Features related to class [Common_Tern] include: Yellow(orange) legs	Features related to class [Caspian_Tern] include: Black legs Yellow(orange) bill
The absence of the following feature No 2, contribute 45.99 % of the difference Description: Yellow(orange) legs 	The presence of the following feature No 1, contribute 32.08 % of the difference Description: Black legs 
Task:	The presence of the following feature No 5, contribute 13.37 % of the difference Description: Yellow(orange) bill 

Question:

**The model predicts the images below as [Caspian_Tern].
Do you think the model made the predictions correctly?**



Fig. 2. A demonstration of the decision-making task (Q1) and the corresponding global explanation. The left instance is correctly classified, while the right one is incorrect. Based on the explanation, the model relies on “leg color” (Concepts 1 & 3). The absence of legs in the right image leads to misclassification. FCCE helps users identify this failure mode by revealing the model’s specific reliance on detected features.

4.2. Task and dataset

To operationalize the hypotheses in a concrete decision-making scenario, we defined a *Model Failure Identification* task. Participants were shown an input image together with the model’s prediction and were asked to determine whether the prediction was correct based on the provided explanation. As illustrated in Fig. 2, successfully completing the task requires understanding the model’s decision logic rather than recognizing the object category itself. For example, identifying that the model relies on leg-related features allows participants to anticipate failure cases when such features are absent or occluded.

We conducted the study on the CUB-200-2011 bird dataset [45]. We selected CUB over large-scale benchmarks such as ImageNet for three reasons. First, CUB classes are semantically well-defined with minimal category overlap. Second, images typically contain a single dominant object, which reduces ambiguity in human judgment. Third, fine-grained bird species recognition is challenging for non-expert users, making the benefits of AI assistance and interpretable explanations more pronounced than in trivial object recognition tasks. Correctness was defined with respect to the ground-truth class label provided by the CUB dataset.

4.3. Experimental conditions

The independent variable in our study was the explanation method presented to participants. We employed a between-subject design and organized the experimental conditions into two complementary sets, each corresponding to a specific hypothesis. Participants were randomly assigned to one experimental condition and completed the same number of tasks under that condition.

4.3.1. Set 1: Comparative analysis (testing H1)

The first set of conditions evaluates the interpretability of FCCE relative to representative state-of-the-art explanation methods. We compared four methods to isolate the effects of concept representation and counterfactual reasoning.

- **CVE [11]** (Non-concept, Counterfactual): Provides visual counterfactual explanations via region replacement. To ensure fairness, we implemented CVE to display all edits required to cross the decision boundary rather than only the single most salient modification.
- **ACE [27]** (Segment-based Concept, Factual): Uses clustered image segments as interpretable concepts.

- **ICE** [15] (Latent Concept, Factual): Learns latent concepts from feature maps without counterfactual information. Including ICE allows us to assess the effect of counterfactual reasoning while holding the concept representation constant.
- **FCCE** (Latent Concept, Counterfactual): Our proposed method.

We excluded CoCoX [14] from the comparison because its explanation quality depends on manual concept filtering to remove noisy segments. In preliminary experiments without such manual curation, the resulting explanations were often unstable, making it difficult to conduct a controlled large-scale human evaluation.

Furthermore, while generative counterfactual methods (e.g., diffusion-based approaches) have recently gained attention, we excluded them from this human study for two principled reasons. First, prior research highlights that pixel-space generation can prioritize visual plausibility over faithfulness to the classifier’s decision boundary, risking the introduction of semantic hallucinations [40,46]. Our task requires users to infer stable, class-level decision rules rather than interpret instance-specific visual edits. Such confounding visual details may mislead users’ generalization and foster unjustified trust. This concern is supported by our own empirical findings with the visual editing baseline (CVE), where participants exhibited an overconfidence effect (high confidence but low accuracy) due to reliance on visual plausibility rather than decision logic. Second, current diffusion-based image generators incur multi-second latency even with GPU acceleration (as detailed in Section 5), making them incompatible with the real-time, interactive decision-support paradigm targeted by FCCE.

We therefore view generative counterfactuals as addressing a complementary objective—high-fidelity post-hoc visualization—rather than serving as directly comparable baselines for real-time interactive decision support. Generative methods excel at post-hoc visualization for inspection, whereas FCCE focuses on extracting abstract, stable concept rules for logic verification. While we exclude generative baselines from this specific decision-support study due to the aforementioned latency and cognitive constraints, future work could explore hybrid interfaces that combine FCCE’s logic with generative visualizations when latency permits.

4.3.2. Set 2: Information ablation (testing H2)

The second set of conditions investigates the impact of explanation information density on human decision-making. Focusing on ICE and FCCE, we systematically varied the amount and type of information presented along two factors.

- **Explanation locality:** Whether only global (class-level) explanations were shown or additional local (instance-level) explanations were provided for the specific test image.
- **Feature labels:** Whether textual descriptions were included to explicitly name the visual concepts. Concept labels were annotated by experts using standard birding guides.

This resulted in eight experimental subgroups (2 methods $\times$ 2 locality settings $\times$ 2 labeling settings). The Global-only, No-label conditions overlap with Set 1 and serve as a shared baseline across both analyzes.

4.4. Experimental procedure

The human behavioral study followed a structured three-phase procedure designed to ensure participant comprehension, data quality, and reliable evaluation of both objective performance and subjective perception.

4.4.1. Phase 1: Training and screening

Participants were first introduced to the task through guided demonstrations illustrating how to interpret the provided explanations and how to assess model predictions. To ensure data quality and participant understanding, Phase 1 concluded with a **catch trial** involving an easily distinguishable class pair with clear visual differences. Only participants who achieved 100% accuracy on the catch trial were allowed to proceed to the main study. This screening step served as a strict attention and comprehension check, filtering out participants who failed to grasp the task or did not engage seriously.

4.4.2. Phase 2: Testing

Participants completed 16 decision-making tasks distributed across four survey pages. Each page focused on a specific pair of visually similar bird classes.

Class selection. We carefully selected class pairs that satisfied three criteria. First, the classes were frequently confused by the CNN, corresponding to low model accuracy. Second, the classes were visually similar and thus challenging for non-expert participants. Third, the classes were distinguishable based on observable visual features, excluding pairs defined primarily by non-visual attributes such as habitat or geographic location.

In addition, for the selected survey instances, we excluded class pairs where the ICE surrogate prediction disagreed with the original CNN prediction, ensuring local consistency between the explanation model and the original classifier. We also ensured that each class appeared in only one pair, and selected task instances with the highest positive scores and lowest negative scores.

For each task, participants were shown an image, the model’s predicted label, and the corresponding explanation. They were then asked to answer three questions:

Table 1

Survey results for **H1**. Best performances and significant omnibus test results (one-way ANOVA, $p < .05$) are highlighted. FCCE achieves the highest objective accuracy (**Q1**) and subjective understanding. Pairwise post-hoc results are reported in the [Appendix B](#).

	Q0	Q1	Q2	Quality			
Groups	Understand	Prediction	Confidence	Understand	Satisfaction	Sufficiency	Completeness
Comparisons (with std deviation) for CVE, ACE, ICE, and FCCE with ANOVA test p values							
CVE	N/A	50.89±8.08	4.02±0.56	3.57±1.09	3.07±1.21	2.93±1.21	2.86±1.29
ACE	3.55±0.89	68.47±11.49	3.83±0.81	3.77±1.02	3.59±1.18	3.09±1.27	3.32±1.32
ICE	4.11±0.70	82.95±13.39	4.16±0.34	4.41±0.59	3.68±0.95	3.32±0.95	3.23±0.81
FCCE	4.42±0.58	89.88±9.58	4.42±0.46	4.43±0.68	4.00±0.95	3.95±0.97	3.81±1.08
ANOVA	p<0.001	p<0.001	p=0.010	p=0.003	p=0.100	p=0.028	p=0.102

- **Q1 (Prediction):** “Do you think the model will predict this image correctly?” (objective accuracy).
- **Q2 (Confidence):** “To what degree are you confident with your choice?” (five-point Likert scale).
- **Q3 (Justification):** “Briefly explain the reason for your choice.”

Responses to **Q3** were used to identify and filter low-quality or nonsensical answers. In addition, participants rated their understanding of the global explanation (**Q0**) at the beginning of each survey page.

4.4.3. Phase 3: Satisfaction

After completing the decision-making tasks, participants rated the explanations they had received along multiple subjective dimensions. Specifically, usefulness, sufficiency, appropriate level of detail, and completeness were measured using established evaluation scales from prior work [47]. This phase captured participants’ overall subjective perception of the explanations, complementing the objective performance metrics collected in Phase 2.

4.5. Participants and implementation

Participants and platform. The survey was designed in Qualtrics and deployed on the Amazon Mechanical Turk platform [48]. We recruited people rated as Masters on the platform and a total of 248 participants passed the catch trials. Participants who failed the catch trial were excluded from the main study but still received the training compensation. Among those who proceeded to the main study, 51 were further excluded from the final analysis because their justifications for **Q3** were meaningless or repeated across multiple answers, leaving 197 participants in the final dataset (approximately 19.7 ± 3.6 per group). A power analysis based on a pilot study suggested a minimum sample size of 18 participants per group. Participants were advised to spend approximately 15 minutes on the training phase (Phase 1) and 35 minutes on the test phases (Phases 2 and 3). We compensated participants with \$5 USD for completing the training phase and \$12 USD for completing the test phases. Participants who failed the catch trial were still paid the \$5 USD training compensation. Participants who proceeded to the test phases could additionally receive a performance-based bonus of up to \$4 USD (\$0.25 USD per task) based on their accuracy. We obtained ethics approval from The University of Melbourne Human Research Ethics Committee (ID 1749428).

Implementation details. All explanations were generated using a ResNet-50 model trained on the CUB dataset, which achieved a top-1 test error of 15.81%. Explanations were computed with respect to features extracted from the final convolutional block (Layer4).

For concept-based methods, we set the number of concepts to $k = 8$ for ICE and FCCE. For ACE, we increased the number of concepts to $k = 20$ to compensate for the increased difficulty of segment-based concept discovery on the CUB dataset. This setting provides ACE with a favorable configuration and ensures a fair comparison.

For FCCE, we used a single class centroid ($k_{\text{cent}} = 1$) and applied sparsity filtering to display only concepts contributing to the first 90% of the prediction score, which typically resulted in 1–3 concepts per explanation. For CVE, we generated a sufficient number of edits so that participants could complete the task effectively, resulting in an average of 3.82 edits per instance in the human-study interface.

4.6. Results and analysis

Results are summarized in [Table 1](#) and [Table 2](#). We report mean values with standard deviations and conducted one-way ANOVA tests ($\alpha = 0.05$), followed by Tukey HSD post-hoc analysis where appropriate.

We emphasize that the goal of this human study is not to evaluate the visual realism of counterfactual instances, but to assess how different explanation paradigms support human decision-making about model reliability and failure cases. Accordingly, all evaluated methods are compared under the same decision-support task, where the explanation must help participants reason about the model’s internal decision logic rather than recognize object categories from images.

Table 2

Explanation information experiment (**H2**). Results show that providing additional information (local explanations or labels) generally reduces objective decision accuracy, suggesting cognitive overload.

Local	Feature	Q0	Q1	Q2	Quality			
Expl.	Labels	Understand	Prediction	Confidence	Understand	Satisfaction	Sufficiency	Completeness
Explanation information experiment (with std deviation) for FCCE								
w/o	w/o	4.42±0.57	89.88±9.58	4.42±0.46	4.43±0.68	4.00±0.95	3.95±0.97	3.81±1.08
w/	w/o	4.41±0.76	83.22±12.51	4.43±0.60	4.53±0.77	4.00±0.81	4.00±0.75	4.00±0.88
w/o	w/	4.74±0.40	80.06±10.76	4.40±0.45	4.67±0.48	4.33±0.91	4.14±1.01	4.10±1.09
w/	w/	4.56±0.57	77.56±9.77	4.45±0.41	4.36±0.49	4.14±0.64	4.32±0.65	4.23±0.68
Explanation information experiment (with std deviation) for ICE								
w/o	w/o	4.11±0.70	82.95±13.39	4.16±0.34	4.41±0.59	3.68±0.95	3.32±0.95	3.23±0.81
w/	w/o	4.23±0.56	77.08±12.31	4.22±0.47	3.83±0.94	3.42±1.17	3.17±1.27	3.17±1.34
w/o	w/	4.39±0.62	75.52±11.93	4.34±0.40	4.33±0.76	4.08±1.02	3.79±0.98	3.42±1.14
w/	w/	4.58±0.59	73.13±9.09	4.42±0.51	4.40±0.75	4.20±1.01	4.05±0.89	3.75±1.29

4.6.1. Overview of results

Overall, FCCE achieved the highest objective decision accuracy and consistently strong subjective ratings across evaluation metrics. Concept-based methods substantially outperformed non-concept baselines, and counterfactual explanations provided additional benefits beyond factual concept representations. All subjective ratings were measured on five-point Likert scales, where higher values indicate better perceived quality. Notably, increasing explanation information density did not improve decision accuracy and, in several cases, led to performance degradation.

4.6.2. H1: Interpretability analysis

Observation. As shown in Table 1, participants using FCCE achieved the highest decision accuracy (89.88%), and the overall group effect is significant (one-way ANOVA, $p < .001$). FCCE significantly outperforms CVE and ACE, while its improvement over ICE is smaller and is not significant under standard post-hoc tests, becoming significant only after applying a standard outlier-removal rule (1.5 IQR; Appendix B). Overall, concept-based methods (FCCE and ICE) substantially outperform non-concept methods (CVE and ACE) on objective accuracy.

Interpretation. These results support **H1** and indicate that abstract concept representations are more effective than visual edits or image segments for supporting human decision-making in fine-grained recognition tasks. Moreover, the difference in performance between FCCE and ICE suggests the added value of counterfactual reasoning. While ICE exposes what concepts are present, FCCE explicitly communicates what needs to change to cross the model's decision boundary, enabling users to better anticipate failure cases. Importantly, the high decision accuracy achieved by participants is consistent with FCCE's counterfactual explanations reflecting the underlying model behavior rather than merely artifacts of the linear surrogate. This behavioral evidence aligns with our computational fidelity analysis (Section 5), where FCCE achieves near-zero Relative Logit Gap in concept space and remains below 0.3% when evaluated on the original CNN after reconstruction, indicating that the generated counterfactuals indeed lie close to the true decision boundary. Pairwise comparisons (reported in the Appendix) further confirm that this improvement remains statistically significant after outlier removal ($p = .033$). Finally, the performance gap between FCCE and ICE persists despite FCCE already operating near the upper bound of task accuracy, indicating that the improvement is not merely a ceiling effect but reflects a qualitative difference in how counterfactual information supports reasoning about model behavior.

4.6.3. H2: Impact of information density

Observation. Table 2 shows that providing additional explanation information does not improve, and often reduces, objective decision accuracy. Participants achieved the highest accuracy when only global explanations without feature labels were shown. Adding local (instance-level) explanations or textual feature labels led to consistent decreases in accuracy for both FCCE and ICE.

Interpretation. These findings reject **H2** and suggest a cognitive-overload effect. The task requires reasoning about class-level failure modes, for which global explanations provide a stable and generalizable mental model. In contrast, additional local explanations and textual labels may introduce redundant or instance-specific information that distracts users from the core decision criteria. This result highlights that more explanation is not necessarily better, and that explanation design must carefully balance completeness and usability.

4.6.4. H3: Subjective perception vs. objective performance

Observation. As shown in Table 1, participants using FCCE reported the highest levels of subjective understanding, satisfaction, and decision confidence. However, high confidence did not always correspond to high accuracy. In particular, participants using CVE exhibited near-random performance (50.89% accuracy) while reporting relatively high confidence scores (4.02/5), exceeding those of ACE.

Interpretation. These results partially support **H3** and reveal a critical disconnect between subjective perception and objective performance. Visual counterfactuals can be perceptually persuasive without being decision-theoretically informative, leading to inflated subjective confidence despite poor objective performance. Similarly, in the information ablation study, adding textual feature labels increased subjective satisfaction while decreasing objective accuracy. These findings underscore the importance of quantitatively evaluating explainability methods rather than relying solely on user preference or self-reported confidence [49,50].

5. Computational evaluation

This section evaluates FCCE from two computational perspectives that are critical for interactive decision support: **(RQ1) real-model fidelity analysis** and **(RQ2) computational efficiency**. In particular, we first examine whether the analytical counterfactual derived in the linear concept space remains faithful when mapped back to the original non-linear CNN. We then assess whether FCCE can generate faithful counterfactual explanations in real time, compared with optimization-based baselines and with a representative pretrained source-conditioned diffusion-based generative counterfactual pipeline used as a latency reference.

Specifically, we address the following research questions:

1. **RQ1 (Fidelity):** Does the analytical counterfactual derived in the linear concept space remain faithful when mapped back to the original non-linear CNN model?
2. **RQ2 (Efficiency / Latency):** How does the computational cost of FCCE compare with existing baselines, and how does it compare with the inference latency of a representative diffusion-based generative counterfactual pipeline?

5.1. Experimental setup

Datasets and models. We evaluate on three benchmarks covering different input resolutions and model complexities. For **CUB** [45], we use a ResNet-50 (448×448) which achieves a Top-1 test error of 15.81% and a Fidelity Error of 5.30%. For **MNIST**, we use a modified ResNet-18 (28×28) with a Top-1 test error of 1.51% and a Fidelity Error of 3.30%. For **ImageNet** (ILSVRC2012 [51]), we use a ResNet-50 (224×224) with a Top-1 test error of 23.85% and a Fidelity Error of 5.33%. **Fidelity error** is defined as the disagreement rate between the linear concept surrogate predictions and the original CNN predictions [15]. The consistently low Fidelity Errors across all datasets confirm that our linear concept space effectively preserves the global decision logic of the original non-linear models.

Crucially, these two metrics address different levels of fidelity. **Fidelity error** follows the standard surrogate-level evaluation used in ICE-style concept-space analysis and validates the *global* consistency of the linear concept surrogate across the data distribution. By contrast, the proposed **Relative Logit Gap (R-Gap)** evaluates the *local* fidelity of a reconstructed analytical counterfactual at the original-model decision boundary. Together, they provide complementary evidence of faithfulness at both the representation level and the instance level.

Regarding test sample selection, we adopt distinct strategies to ensure robust evaluation. For CUB, we employ the identical class pairs and instances used in our human study (Section 4) to ensure direct alignment between computational metrics and human behavioral results. For MNIST, we randomly select 5 non-overlapping class pairs. For ImageNet, we select 5 frequently confused class pairs to stress-test the method. In all cases, we use a fixed pairwise evaluation protocol and apply the same counterfactual evaluation procedure once the class pairs and test instances are selected.

Baselines. We compare FCCE against three categories of methods: (1) factual explainers (**ACE**, **ICE**); (2) counterfactual optimization (**CVE**); and (3) a numerical optimization variant of our method (**FCCE (Numerical)**), which serves as an internal baseline to isolate the benefit of the closed-form analytical solution (**FCCE (Analytical)**).

Generative models are another way that counterfactuals could be visualised. To contextualize the runtime cost of generative counterfactual explanations, we additionally report the latency of a representative diffusion-based counterfactual generation pipeline [8]. Following the source-conditioned, target-guided paradigm of prior diffusion-based counterfactual work, we implement a minimal source-conditioned diffusion pipeline for counterfactual generation using the pretrained 256×256 ImageNet Guided Diffusion model [44]. Given a source image and a target class, the source image is first forward-noised to the starting point of the respaced diffusion trajectory, after which classifier-guided DDIM sampling is performed toward the target class.

This implementation is used as a task-related latency reference rather than as a dataset-specific quality baseline. We use the publicly available pretrained ImageNet model without retraining, since the purpose of this experiment is to estimate the practical inference cost of source-conditioned diffusion-based counterfactual generation. We restrict this benchmark to ImageNet because the pretrained diffusion and classifier pair is directly available in this setting, whereas comparable pretrained source-conditioned diffusion counterfactual models are not readily available for CUB or MNIST, and our experiments on ImageNet were clear that diffusion-based models are several orders of magnitude too slow for real-time counterfactual generation. Further, this approach avoids introducing additional dataset-specific training or adaptation overhead that would confound the latency comparison.

Hardware protocol. All non-generative explanation methods are measured on a 16-core CPU to reflect realistic deployment without specialized accelerators. Model inference time is measured on a GTX 1080Ti GPU for reference. Diffusion-based generators require GPU execution; we report their latency on an NVIDIA Tesla T4, a widely used cloud inference GPU.

For reproducibility, all diffusion measurements are taken after one warm-up run and then averaged over repeated executions on the same hardware. Guided Diffusion uses the OpenAI pretrained 256×256 ImageNet model together with its corresponding classifier,

Table 3

Fidelity analysis in concept and real-model spaces. We report the **Relative Logit Gap** (*R-Gap*, %) for the original instance and the corresponding counterfactual. **Original Instance** reflects the baseline relative logit gap of the original prediction. **FCCE (Analytical)** and **FCCE (Numerical)** report the post-counterfactual fidelity. **Concept Space** denotes evaluation in the linear concept space, while **Real Model** denotes evaluation after reconstruction on the original non-linear CNN. Lower values indicate higher fidelity, with 0% corresponding to a solution that effectively lies on the decision boundary. FCCE (Analytical) consistently achieves negligible *R-Gap*, and the real-model results confirm that this fidelity is preserved after mapping back to the original non-linear model.

Dataset	Space	Original Instance	FCCE (Analytical)	FCCE (Numerical)
CUB	Concept Space	26.9730 $\pm$ 26.7863	0.0000 $\pm$ 0.0000	0.0315 $\pm$ 0.0185
	Real Model	26.9691 $\pm$ 26.7920	0.0315 $\pm$ 0.0144	0.0463 $\pm$ 0.0364
MNIST	Concept Space	78.2647 $\pm$ 32.7703	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000
	Real Model	78.2274 $\pm$ 32.7801	0.2544 $\pm$ 0.2299	3.2573 $\pm$ 12.0704
ImageNet	Concept Space	38.5153 $\pm$ 33.4398	0.0000 $\pm$ 0.0000	0.1308 $\pm$ 0.1392
	Real Model	38.5007 $\pm$ 33.4347	0.0583 $\pm$ 0.0276	0.1233 $\pm$ 0.1237

DDIM sampling with 50 steps, and classifier guidance enabled at each denoising step. For each test instance, the source image is first noised to the starting point of the respaced diffusion trajectory, after which reverse sampling is performed under the target-class guidance associated with the class-pair protocol. Accordingly, the diffusion latency reference is reported on the ImageNet setting only. This setup yields a minimal source-conditioned generative counterfactual pipeline whose latency can be directly compared against FCCE in terms of per-instance interaction cost.

5.2. RQ1: Real-model fidelity of analytical counterfactuals

A central concern for FCCE is not only whether the ICE-derived concept space provides a faithful surrogate of the original model, but more specifically whether a counterfactual computed analytically in this linearized space remains valid after reconstruction in the original non-linear CNN. The former has already been assessed through surrogate-level **Fidelity Error** in the experimental setup, which captures the fidelity of the ICE-style surrogate model. The present subsection focuses on the latter, namely *counterfactual fidelity*, which is directly relevant to FCCE.

To evaluate real-model fidelity, we use the Relative Logit Gap (*R-Gap*) defined in Section 3.6. Table 3 reports the corresponding results on CUB, MNIST, and ImageNet.

Results. Across all datasets, **FCCE (Analytical)**—the default instantiation of our method—achieves near-zero *R-Gap* in concept space and remains below 0.3% in the real-model evaluation after reconstruction, despite the counterfactual being computed in a simplified linear concept space. This indicates that the analytical solution remains highly faithful to the true decision boundary of the original non-linear model. Compared to **FCCE (Numerical)**, the analytical solution attains comparable or better fidelity with substantially lower variance, suggesting a more stable and precise boundary solution rather than sensitivity to local minima introduced by iterative search.

These results are particularly important because low surrogate-level **Fidelity Error** alone does not guarantee that an individual reconstructed counterfactual remains valid on the original model. By contrast, the proposed *R-Gap* directly verifies whether the analytical counterfactual preserves decision-boundary faithfulness when evaluated on the original non-linear CNN, which is the more task-relevant notion of fidelity for FCCE.

In addition to real-model fidelity, Table 4 reports standard counterfactual optimization losses. **FCCE (Analytical)** achieves comparable or lower validity, similarity, and plausibility losses than the numerical baseline, confirming that its closed-form efficiency does not come at the expense of counterfactual quality.

5.3. RQ2: Computational efficiency and real-time interaction

For FCCE and non-generative baselines, computational costs are reported across all three datasets. For the diffusion-based latency reference, we report the ImageNet setting only, because the required pretrained diffusion-classifier pair is directly available only in that setting.

Table 5 summarizes computational costs. We report both **Global** and **Local** time, where **Global** denotes one-time offline preprocessing (e.g., concept discovery and class-level explanation precomputation), and **Local** denotes per-instance inference-time overhead during deployment.

Results. **FCCE (Analytical)**, which represents the proposed FCCE framework in its closed-form instantiation, incurs negligible local overhead, typically on the order of 10^{-5} – 10^{-2} seconds, which is comparable to factual explainers and effectively dominated by a single CNN forward pass. In contrast, optimization-based counterfactual methods such as CVE require multiple costly edits per instance and are substantially slower.

Table 4

Counterfactual optimization loss analysis. We report standard counterfactual evaluation losses, including **Validity Loss** (L_{vali}), **Similarity Loss** (L_{simi}), and **Plausibility Loss** (L_{plau}). Lower values indicate better counterfactual quality. FCCE (Analytical) consistently achieves comparable or lower losses than numerical search, indicating that the closed-form solution does not sacrifice optimality while avoiding iterative optimization.

Dataset	Method	L_{vali}	L_{simi}	L_{plau}
CUB	FCCE (Numerical)	$(2.15 \pm 3.83) \times 10^{-4}$	$(5.09 \pm 1.38) \times 10^{-1}$	$(4.30 \pm 1.29) \times 10^{-1}$
	FCCE (Analytical)	$< 1.00 \times 10^{-6}$	$(3.74 \pm 3.89) \times 10^{-1}$	$(3.88 \pm 1.26) \times 10^{-1}$
MNIST	FCCE (Numerical)	$(1.02 \pm 3.40) \times 10^{-5}$	$(9.36 \pm 2.66) \times 10^0$	$(6.52 \pm 2.72) \times 10^0$
	FCCE (Analytical)	$< 1.00 \times 10^{-6}$	$(4.66 \pm 2.72) \times 10^0$	$(3.91 \pm 0.94) \times 10^0$
ImageNet	FCCE (Numerical)	$(2.20 \pm 3.10) \times 10^{-4}$	$(2.94 \pm 0.73) \times 10^{-1}$	$(2.56 \pm 0.78) \times 10^{-1}$
	FCCE (Analytical)	$< 1.00 \times 10^{-6}$	$(1.74 \pm 1.30) \times 10^{-1}$	$(2.21 \pm 0.53) \times 10^{-1}$

Table 5

Computational time cost and interactivity analysis (seconds). We report both **Global** (one-time offline preprocessing) and **Local** (per-instance inference-time) costs. GPU inference time of the backbone CNN is shown for reference. For FCCE and non-generative baselines, results are reported on CUB, MNIST, and ImageNet. We additionally report the latency of a representative pretrained source-conditioned diffusion-based generative counterfactual pipeline on an NVIDIA Tesla T4 GPU under a fixed protocol (256 $\times$ 256, 50 DDIM steps). Specifically, this diffusion reference uses the pretrained ImageNet Guided Diffusion model with classifier guidance, where each source image is first noised to the starting point of the respaced diffusion trajectory and then reverse-sampled toward the target class. The diffusion row is reported separately because it serves as a task-related latency reference in the ImageNet setting only, rather than as a dataset-specific counterfactual quality baseline.

Method	Computational Time Cost (seconds)	
	Global	Local
CUB (ResNet50), GPU Inference: $(1.68 \pm 0.05) \times 10^{-2}$		
ACE	$(6.53 \pm 0.28) \times 10^1$	N/A
ICE	$(2.58 \pm 0.31) \times 10^1$	$(1.15 \pm 0.28) \times 10^{-2}$
CVE (Greedy)	N/A	5.84 edits $\times$ $(9.01 \pm 0.08) \times 10^1$
CVE (Search)	N/A	5.84 edits $\times$ $(1.43 \pm 0.09) \times 10^0$
FCCE (Numerical)	$(2.58 \pm 0.31) \times 10^1$	$(1.48 \pm 0.39) \times 10^{-2}$
FCCE (Analytical)	$(2.58 \pm 0.31) \times 10^1$	$(1.15 \pm 0.28) \times 10^{-2}$
MNIST (ResNet18), GPU Inference: $(3.35 \pm 1.40) \times 10^{-3}$		
ACE	$(1.37 \pm 0.02) \times 10^1$	N/A
ICE	$(1.36 \pm 0.32) \times 10^1$	$(3.72 \pm 0.58) \times 10^{-4}$
CVE (Greedy)	N/A	3.01 edits $\times$ $(2.72 \pm 0.22) \times 10^{-1}$
CVE (Search)	N/A	3.01 edits $\times$ $(5.15 \pm 1.97) \times 10^{-2}$
FCCE (Numerical)	$(1.36 \pm 0.32) \times 10^1$	$(1.46 \pm 0.64) \times 10^{-2}$
FCCE (Analytical)	$(1.36 \pm 0.32) \times 10^1$	$(3.72 \pm 0.58) \times 10^{-4}$
ImageNet (ResNet50), GPU Inference: $(1.70 \pm 0.14) \times 10^{-2}$		
ACE	$(3.14 \pm 0.17) \times 10^1$	N/A
ICE	$(8.37 \pm 0.18) \times 10^1$	$(9.82 \pm 1.19) \times 10^{-3}$
CVE (Greedy)	N/A	4.73 edits $\times$ $(4.39 \pm 0.08) \times 10^0$
CVE (Search)	N/A	4.73 edits $\times$ $(3.03 \pm 0.03) \times 10^{-1}$
FCCE (Numerical)	$(8.37 \pm 0.18) \times 10^1$	$(1.47 \pm 0.41) \times 10^{-2}$
FCCE (Analytical)	$(8.37 \pm 0.18) \times 10^1$	$(9.82 \pm 1.19) \times 10^{-3}$
ImageNet diffusion latency reference, 256 $\times$ 256, 50 DDIM steps (T4 GPU)		
Source-conditioned Guided Diffusion	N/A	16.34 ± 0.46

The diffusion latency reference further highlights the gap between diffusion-based generative counterfactual pipelines and real-time interactivity. Under the source-conditioned 256 $\times$ 256 50-step protocol in the ImageNet setting, the minimal Guided Diffusion pipeline requires 16.34 ± 0.46 seconds per image. Even as a pretrained GPU-based pipeline without dataset-specific retraining, it remains orders of magnitude slower than **FCCE (Analytical)** running on CPU. This result is particularly notable because the diffusion reference is evaluated in its native pretrained setting, without the additional adaptation cost that would typically be required for other datasets.

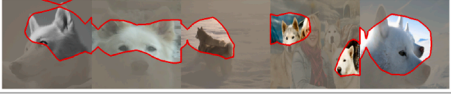
Class [Eskimo dog, husky] is different from Class [malamute, malemute, Alaskan malamute] with around 15.09% difference. The absence/presence of the following 4 features make up 98.49% of this 15.09% difference, support the instances in Class [Eskimo dog, husky].	
Features related to class [malamute, malemute, Alaskan malamute]	Features related to class [Eskimo dog, husky]
The absence of the following feature No 3, contribute 50.22 % of the difference 	The presence of the following feature No 0, contribute 40.18 % of the difference 
The absence of the following feature No 8, contribute 4.34 % of the difference 	The presence of the following feature No 9, contribute 3.75 % of the difference 

Fig. 3. Global counterfactual explanation for Husky vs. Malamute. Beyond commonly cited facial features, FCCE identifies Tail Curl (Feature 8) as a discriminative factor, revealing a non-standard but valid visual correlation learned by the model.

Overall, **FCCE (Analytical)** achieves orders of magnitude speedup over optimization-based methods in end-to-end explanation latency, and over the diffusion latency reference in per-instance generation time, enabling real-time interaction without specialized hardware.

5.4. Summary

Taken together, these results answer both research questions. **FCCE (Analytical)** produces counterfactual explanations that are **faithful** to the original CNN decision boundary (**RQ1**), while simultaneously achieving **real-time** performance with negligible computational overhead (**RQ2**). This combination distinguishes FCCE from both search-based methods and the diffusion latency reference, both of which involve substantially higher latency.

6. Case studies and examples

In this section, we present qualitative examples to demonstrate the discovery and debugging capabilities of FCCE across different domains. These examples complement our quantitative evaluation by illustrating how concept-based counterfactual explanations reveal model decision cues in practice.

Discovering non-standard features (Husky vs. Malamute). Fig. 3 reveals an interesting model behavior: while common descriptions distinguish Huskies and Malamutes by eye color or face patterns, FCCE indicates that the CNN relies on **Tail Curl**. Malamutes often have curled tails, while Huskies typically do not. Although this feature is not always emphasized as a primary diagnostic criterion, it represents a valid visual correlation captured by the model. This example highlights FCCE’s ability to uncover non-obvious but semantically meaningful decision cues.

Validating decision logic (Swallows). Fig. 4 shows the FCCE for a binary classifier of two swallow species. Key visual differences include forehead and neck/belly appearance. FCCE highlights Features 5 and 0 (forehead-related regions) and Feature 2 (neck/belly area), confirming that the model bases its decision on semantically appropriate features. This qualitative observation is consistent with our real-model fidelity analysis in Section 5.2, which shows that FCCE counterfactuals remain faithful after mapping back to the original CNN.

Debugging model shortcuts (Barbershop vs. Bookshop). Fig. 5 shows a model misclassifying a bookshop as a barbershop. FCCE explains why: the model over-associates storefront signs with the barbershop class, and thus incorrectly treats the “Amazon Books” sign as evidence for barbershops. This semantic-level debugging is notoriously difficult with standard pixel-level heatmaps, which might highlight the sign as “important” but fail to explicitly link it to the “barbershop” class decision logic. FCCE bridges this gap by identifying the specific concept driving the misclassification.

Medical domain application (Skin cancer). Fig. 6 shows an explanation for the HAM10000 skin cancer dataset [52]. The extracted concepts correspond to clinically meaningful dermoscopic patterns (e.g., leaf-like areas and dotted vessels). Informal feedback from experienced dermatologists suggests that these features are consistent with standard dermoscopic criteria. This example illustrates the potential of FCCE to provide rapid visual summaries that support expert verification in safety-critical domains.

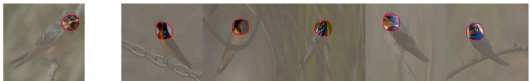
The following image is predicted as [Barn_Swallow] but not [Cliff_Swallow] with around 30.16% difference. The absence/presence of the following 3 features make up 93.12% of this 30.16% difference, support the image in Class [Barn_Swallow]. 	
Features related to class [Cliff_Swallow] The absence of the following feature No 5, contribute 12.40 % of the difference 	Features related to class [Barn_Swallow] The presence of the following feature No 2, contribute 56.12 % of the difference 
	The presence of the following feature No 0, contribute 24.61 % of the difference 

Fig. 4. Local explanation for a CUB class pair. The highlighted concepts correspond to biologically meaningful regions (forehead, neck, and belly), indicating that the model relies on appropriate semantic cues.

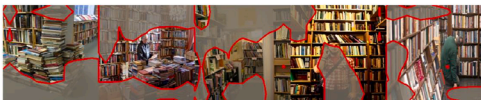
The following image is predicted as [barbershop] but not [bookshop, bookstore, bookstall] with around 31.87% difference. The absence/presence of the following 4 features make up 98.42% of this 31.87% difference, support the image in Class [barbershop]. 	
Features related to class [bookshop, bookstore, bookstall] The absence of the following feature No 1, contribute 62.39 % of the difference 	Features related to class [barbershop] The presence of the following feature No 2, contribute 8.28 % of the difference 
The absence of the following feature No 0, contribute 23.08 % of the difference 	The presence of the following feature No 6, contribute 4.67 % of the difference 

Fig. 5. Global counterfactual explanation exposing a model shortcut. Storefront signs are strongly associated with barbershops in the training data, leading the model to misinterpret the “Amazon Books” sign as evidence for the barbershop class.

Class [0_akiec] is different from Class [1_bcc] with around 252.60% difference. The absence/presence of the following 5 features make up 88.92% of this 252.60% difference, support the instances in Class [0_akiec].	
Features related to class [1_bcc]	Features related to class [0_akiec]
The absence of the following feature No 3, contribute 21.65 % of the difference	The presence of the following feature No 6, contribute 22.74 % of the difference
The absence of the following feature No 5, contribute 14.13 % of the difference	The presence of the following feature No 1, contribute 15.59 % of the difference
	The presence of the following feature No 4, contribute 14.81 % of the difference

Fig. 6. FCCE explanation for the HAM10000 skin cancer dataset. Extracted concepts correspond to clinically meaningful dermoscopic patterns, illustrating the applicability of FCCE in medical imaging.

7. Discussion and conclusion

In this work, we presented **Fast Concept-based Counterfactual Explanations (FCCE)**, a framework designed to bridge the gap between semantic interpretability and real-time interactivity. By shifting counterfactual reasoning from the high-dimensional pixel space to a linearized concept space, FCCE derives a closed-form analytical solution that enables real-time decision support.

Implications for explainable AI. Our findings offer three key implications for the design of interpretable systems:

1. **Human-centric effectiveness.** Our large-scale human study ($N = 197$) provides strong empirical evidence that concept-based counterfactuals significantly improve human decision-making regarding model reliability (89.88% accuracy), outperforming both feature-attribution and pixel-level counterfactual baselines. Furthermore, our results challenge the assumption that “more information is always better.” Participants performed best with concise, global concept rules, supporting a “less-is-more” principle for mitigating cognitive overload in XAI.
2. **Real-time interactivity.** Practical deployment of XAI requires low latency. By deriving a closed-form analytical solution, FCCE computes the analytical counterfactual update itself in under 10^{-5} seconds, while the end-to-end local explanation overhead remains on the order of 10^{-2} seconds. This represents an orders-of-magnitude speedup compared to optimization-based methods and the diffusion latency reference. FCCE demonstrates that real-time interactivity is achievable without specialized hardware, making it suitable for standard CPU deployment.
3. **Fidelity of linear concepts.** A common concern with concept-based explanations is the fidelity of the linear surrogate. Our computational evaluation addresses this directly: FCCE counterfactuals remain highly faithful when mapped back to the original non-linear CNN, achieving a **Relative Logit Gap below 0.3%** in real-model evaluation. This confirms that the locally linear concept-space construction used by FCCE provides a highly faithful approximation for counterfactual reasoning near the decision boundary of the original model.

Limitations and future work. Our approach relies on the quality of the discovered concept space. While unsupervised matrix factorization (ICE) avoids annotation costs, the resulting concepts may occasionally lack semantic distinctiveness compared to supervised approaches. Moreover, standard NMF enforces non-negative activations, which may limit the representation of inhibitory features. Future work could explore semi-supervised concept discovery or integrate semi-NMF approaches (e.g., Archetypal SAE [35]) to capture these features while retaining our analytical efficiency. Additionally, while our linearity assumption holds well for ResNet architectures as evidenced by our fidelity analysis, highly non-linear decision manifolds in other architectures (e.g., Vision Transformers) may require piecewise-linear or kernelized extensions. Finally, extending FCCE to support human-in-the-loop debugging, where users can interactively edit concepts to retrain the model, remains a promising direction for reliable AI deployment.

CRedit authorship contribution statement

Ruihan Zhang: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Data curation, Conceptualization; **Tim Miller:** Writing – review & editing, Validation, Supervision, Funding acquisition, Conceptualization; **Krista A. Ehinger:** Writing – review & editing, Validation, Supervision, Conceptualization; **Benjamin I. P. Rubinstein:** Writing – review & editing, Validation, Supervision, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Co-author was a guest editor in AIJ. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Part of this work was conducted while Ruihan Zhang was a PhD student at The University of Melbourne.

Data availability

Github link to data and source code is available in the article.

Appendix A. Human study samples

Figs. A.7–A.10 illustrate the graphical user interface and questions presented to participants during the survey. Detailed survey materials and the complete interface code are available in the supplementary repository: https://github.com/zhangrh93/FCCE_survey.

Question 0

To what degree do you **understand** the given global explanation?

You need to score from 1 to 5. 1 means not understandable, 5 means fully understandable.

1	2	3	4	5
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Fig. A.7. Example of Q0 (Understanding Check) in the human survey.

Question 1

Based on what you learned from the explanations, Do you think the AI robot has made the correct prediction for this image?

Hint: The AI robot make predictions based on the identified concepts. If it made mistakes when identifying the concepts, it may provide a wrong prediction.

You can analysis the given image and the explanation to find out if the AI robot may make mistakes idenfying the concepts in the image. Common mistakes include:

1. Important concepts are too small or even missed in the given images.
2. The AI robot idenfied a concept with a wrong region in the image.
3. The AI robot confused a concept in the image with another concept.

Yes, the AI robot classified this image correctly.

No, the AI robot made a mistake here.

Fig. A.8. Example of Q1 (Prediction Task) in the human survey.

Question 2

To what degree you are confident with your choice to the previous question.

You need to select a score from 1 to 5. 1 means not confident at all, 5 means fully confident with your choice.

1

2

3

4

5

Fig. A.9. Example of Q2 (Confidence Rating) in the human survey.

Question 3

Please explain the reason for your choices.

Which feature or features lead to this results? Does the AI robot identify these features correctly?

Please directly use **the feature numbers** in the explanations to support your reasons.

Fig. A.10. Example of Q3 (Justification) in the human survey.

Appendix B. Statistical analysis of human study

This section provides detailed statistical results supporting the findings in [Section 4](#).

B.1. Pairwise comparisons for hypothesis H1

Following the significant omnibus ANOVA result ($F(3, 76) = 17.52, p < .001$), we conducted Tukey HSD post-hoc tests to determine specific group differences. The results are visualized in [Fig. B.11](#).

Multiple Comparisons							
Tukey HSD							
Dependent Variable	(I) sgid	(J) sgid	Mean Difference (I-J)	Std. Error	Sig.	95% Confidence Interval	
						Lower Bound	Upper Bound
Q1_Prediction	ACEglo	CVEloc	.1757305*	.0379025	<.001	.076139	.275322
		FCCEglo	-.2141504*	.0338224	<.001	-.303022	-.125279
		ICEglo	-.1448864*	.0334268	<.001	-.232718	-.057055
	CVEloc	ACEglo	-.1757305*	.0379025	<.001	-.275322	-.076139
		FCCEglo	-.3898810*	.0382518	<.001	-.490391	-.289371
		ICEglo	-.3206169*	.0379025	<.001	-.420209	-.221025
	FCCEglo	ACEglo	.2141504*	.0338224	<.001	.125279	.303022
		CVEloc	.3898810*	.0382518	<.001	.289371	.490391
		ICEglo	.0692641	.0338224	.180	-.019607	.158135
	ICEglo	ACEglo	.1448864*	.0334268	<.001	.057055	.232718
		CVEloc	.3206169*	.0379025	<.001	.221025	.420209
		FCCEglo	-.0692641	.0338224	.180	-.158135	.019607
Q2_Confidence	ACEglo	CVEloc	-.1956169	.1955259	.750	-.709377	.318143
		FCCEglo	-.5959145*	.1744784	.006	-1.054370	-.137459
		ICEglo	-.3380682	.1724377	.212	-.791162	.115026
	CVEloc	ACEglo	.1956169	.1955259	.750	-.318143	.709377
		FCCEglo	-.4002976	.1973280	.187	-.918793	.118198
		ICEglo	-.1424513	.1955259	.885	-.656211	.371309
	FCCEglo	ACEglo	.5959145*	.1744784	.006	.137459	1.054370
		CVEloc	.4002976	.1973280	.187	-.118198	.918793
		ICEglo	.2578463	.1744784	.456	-.200610	.716302
	ICEglo	ACEglo	.3380682	.1724377	.212	-.115026	.791162
		CVEloc	.1424513	.1955259	.885	-.371309	.656211
		FCCEglo	-.2578463	.1744784	.456	-.716302	.200610
Qua_Understand	ACEglo	CVEloc	.201	.289	.899	-.56	.96
		FCCEglo	-.656	.258	.062	-1.33	.02
		ICEglo	-.636	.255	.069	-1.31	.03
	CVEloc	ACEglo	-.201	.289	.899	-.96	.56
		FCCEglo	-.857*	.292	.023	-1.62	-.09
		ICEglo	-.838*	.289	.025	-1.60	-.08
	FCCEglo	ACEglo	.656	.258	.062	-.02	1.33
		CVEloc	.857*	.292	.023	.09	1.62
		ICEglo	.019	.258	1.000	-.66	.70
	ICEglo	ACEglo	.636	.255	.069	-.03	1.31
		CVEloc	.838*	.289	.025	.08	1.60
		FCCEglo	-.019	.258	1.000	-.70	.66
Qua_Sufficiency	ACEglo	CVEloc	.162	.375	.973	-.82	1.15
		FCCEglo	-.861	.335	.057	-1.74	.02
		ICEglo	-.227	.331	.902	-1.10	.64
	CVEloc	ACEglo	-.162	.375	.973	-1.15	.82
		FCCEglo	-1.024*	.379	.042	-2.02	-.03
		ICEglo	-.390	.375	.728	-1.38	.60
	FCCEglo	ACEglo	.861	.335	.057	-.02	1.74
		CVEloc	1.024*	.379	.042	.03	2.02
		ICEglo	.634	.335	.240	-.25	1.51
	ICEglo	ACEglo	.227	.331	.902	-.64	1.10
		CVEloc	.390	.375	.728	-.60	1.38
		FCCEglo	-.634	.335	.240	-1.51	.25

*. The mean difference is significant at the 0.05 level.

Fig. B.11. Post-hoc Tukey HSD test results for Hypothesis H1. The table details pairwise mean differences and confidence intervals.

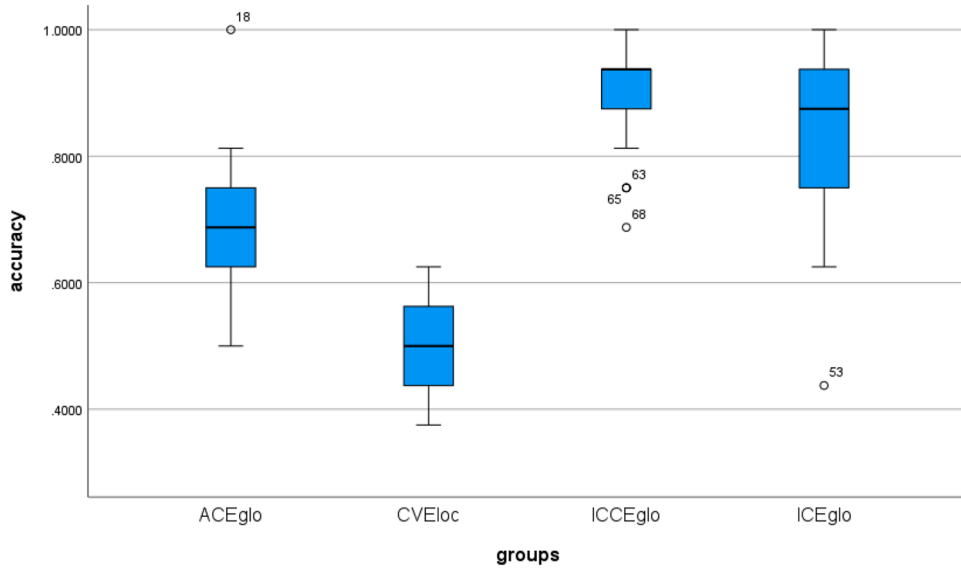


Fig. B.12. Box plot of participant decision accuracy across the four method groups. Note the tighter distribution for FCCE compared to baselines.

Analysis of ceiling effects and outliers. As shown in Table 1, the comparison between FCCE (89.88%) and ICE (82.95%) yielded a p-value of $p = .180$ in the initial analysis. The box plot in Fig. B.12 reveals that the distribution of FCCE scores is tightly clustered near the upper bound (1.0), indicating a potential ceiling effect typical in high-performance assistance tasks. Furthermore, the presence of outliers in the lower quartiles suggests that a subset of participants may have disengaged or misunderstood the task despite passing the screening. To address this, we performed a secondary robust analysis by excluding outliers defined by the standard $1.5 \times \text{IQR}$ rule. Under this outlier-removed condition, the performance difference between FCCE and ICE becomes statistically significant ($p = .033$), confirming that for the majority of attentive participants, FCCE provides a measurable advantage over the factual baseline.

B.2. Factorial analysis for hypothesis H2

To evaluate Hypothesis H2 (Information Density), we conducted a $2 \times 2 \times 2$ factorial ANOVA targeting the method (FCCE/ICE), explanation locality (Global/Local), and feature labeling (With/Without).

Objective accuracy (Q1). As shown in Fig. B.13, all three main factors showed significant main effects on objective accuracy. No significant 2-way or 3-way interaction effects were observed, suggesting that the cognitive load induced by additional information (local explanations or text labels) acts additively to reduce decision performance in this rapid identification task.

Subjective metrics. Fig. B.14 presents the factorial analysis for subjective ratings. Consistent with the objective results, the “Explanation Type” (Local vs. Global) was not a significant factor for subjective satisfaction, reinforcing the finding that global concept rules alone are sufficient and preferred by users for this task.

Tests of Between-Subjects Effects						
Dependent Variable: Q1_Prediction						
Source	Type III Sum of Squares	df	Mean Square	F	Sig.	Partial Eta Squared
Corrected Model	.410 ^a	7	.059	4.657	<.001	.176
Intercept	98.806	1	98.806	7855.106	<.001	.981
factual_counterfactual	.117	1	.117	9.331	.003	.057
explanation_type	.073	1	.073	5.835	.017	.037
with_text	.175	1	.175	13.883	<.001	.083
factual_counterfactual * explanation_type	.000	1	.000	.015	.902	.000
factual_counterfactual * with_text	.004	1	.004	.322	.571	.002
explanation_type * with_text	.014	1	.014	1.119	.292	.007
factual_counterfactual * explanation_type * with_text	.000	1	.000	.009	.925	.000
Error	1.925	153	.013			
Total	105.395	161				
Corrected Total	2.335	160				

a. R Squared = .176 (Adjusted R Squared = .138)

Fig. B.13. Factorial ANOVA results for objective prediction accuracy (Q1). Significant p-values indicate that adding local information or labels significantly impacts performance.

C. Computational implementation details

To ensure reproducibility of the computational evaluation presented in Section 5, we provide detailed parameter settings for all methods and baselines.

C.1. Model and data preprocessing

- **CUB-200-2011:** Images were resized to 448×448 . We used a ResNet-50 backbone fine-tuned on the dataset. Concepts were extracted from the final convolutional layer (layer4).
- **MNIST:** Images were resized to 28×28 . We used a modified ResNet-18 suited for low-resolution input.
- **ImageNet:** Images were center-cropped to 224×224 . We used the standard pre-trained ResNet-50 provided by PyTorch.

C.2. FCCE configuration

For the analytical solution, no hyperparameters are required for the optimization itself. For the concept extraction step (via ICE [15]):

- **Number of Concepts (k):** Set to $k = 8$ for CUB/ImageNet and $k = 5$ for MNIST to capture dominant semantic factors.
- **Sparsity:** For display, we retained concepts contributing to the top 90% of the decision score change, typically resulting in 1–3 active concepts per explanation.

C.3. Baseline configurations

All baselines were tuned to reflect standard usage scenarios. The diffusion-based latency reference was evaluated on an NVIDIA Tesla T4 GPU.

CVE (Counterfactual visual explanations) [11]. The candidate pool size was set to $N = 30$ images per class. We used a greedy search strategy to identify the minimal set of region swaps required to flip the prediction.

Source-conditioned guided diffusion (Latency reference). We used the OpenAI pretrained 256×256 ImageNet guided-diffusion model together with the corresponding classifier for classifier guidance [44]. Following the source-conditioned, target-guided paradigm used in prior diffusion-based counterfactual generation work [8], each run takes as input a source image and a target class.

For the diffusion model, we used the standard ImageNet 256×256 configuration with class conditioning enabled, 1000 diffusion steps, linear noise schedule, 256 channels, 2 residual blocks, and attention resolutions at 32, 16, 8. Sampling was accelerated with DDIM using a respaced 50-step trajectory. The diffusion model was executed in `float16` precision.

Tests of Between-Subjects Effects

Dependent Variable: Q0_Understand

Source	Type III Sum of Squares	df	Mean Square	F	Sig.	Partial Eta Squared
Corrected Model	5.508 ^a	7	.787	2.158	.041	.090
Intercept	3032.459	1	3032.459	8318.225	<.001	.982
factual_counterfactual	1.610	1	1.610	4.417	.037	.028
explanation_type	.032	1	.032	.088	.767	.001
with_text	2.861	1	2.861	7.847	.006	.049
factual_counterfactual * explanation_type	.593	1	.593	1.625	.204	.011
factual_counterfactual * with_text	.052	1	.052	.144	.705	.001
explanation_type * with_text	.023	1	.023	.064	.800	.000
factual_counterfactual * explanation_type * with_text	.147	1	.147	.403	.526	.003
Error	55.777	153	.365			
Total	3229.938	161				
Corrected Total	61.285	160				

a. R Squared = .090 (Adjusted R Squared = .048)

(a) Understanding (Q0)

Tests of Between-Subjects Effects

Dependent Variable: Q2_Confidence

Source	Type III Sum of Squares	df	Mean Square	F	Sig.	Partial Eta Squared
Corrected Model	1.512 ^a	7	.216	1.043	.403	.046
Intercept	2934.226	1	2934.226	14172.044	<.001	.989
factual_counterfactual	.720	1	.720	3.479	.064	.022
explanation_type	.092	1	.092	.443	.507	.003
with_text	.324	1	.324	1.566	.213	.010
factual_counterfactual * explanation_type	.020	1	.020	.095	.758	.001
factual_counterfactual * with_text	.344	1	.344	1.660	.200	.011
explanation_type * with_text	.011	1	.011	.051	.821	.000
factual_counterfactual * explanation_type * with_text	.001	1	.001	.004	.951	.000
Error	31.678	153	.207			
Total	3094.629	161				
Corrected Total	33.189	160				

a. R Squared = .046 (Adjusted R Squared = .002)

(b) Confidence (Q2)

Tests of Between-Subjects Effects

Dependent Variable: Qua_Understand

Source	Type III Sum of Squares	df	Mean Square	F	Sig.	Partial Eta Squared
Corrected Model	5.804 ^a	7	.829	1.793	.092	.076
Intercept	2953.916	1	2953.916	6387.482	<.001	.977
factual_counterfactual	2.463	1	2.463	5.325	.022	.034
explanation_type	1.233	1	1.233	2.667	.105	.017
with_text	.775	1	.775	1.676	.197	.011
factual_counterfactual * explanation_type	.223	1	.223	.482	.488	.003
factual_counterfactual * with_text	.417	1	.417	.902	.344	.006
explanation_type * with_text	.141	1	.141	.305	.581	.002
factual_counterfactual * explanation_type * with_text	2.630	1	2.630	5.687	.018	.036
Error	70.755	153	.462			
Total	3190.000	161				
Corrected Total	76.559	160				

a. R Squared = .076 (Adjusted R Squared = .034)

(c) Understanding Quality

Tests of Between-Subjects Effects

Dependent Variable: Qua_Satisfaction

Source	Type III Sum of Squares	df	Mean Square	F	Sig.	Partial Eta Squared
Corrected Model	9.995 ^a	7	1.428	1.655	.124	.070
Intercept	2451.839	1	2451.839	2842.329	<.001	.949
factual_counterfactual	2.860	1	2.860	3.316	.071	.021
explanation_type	.288	1	.288	.334	.564	.002
with_text	6.616	1	6.616	7.670	.006	.048
factual_counterfactual * explanation_type	.006	1	.006	.007	.935	.000
factual_counterfactual * with_text	1.236	1	1.236	1.433	.233	.009
explanation_type * with_text	.083	1	.083	.096	.757	.001
factual_counterfactual * explanation_type * with_text	.810	1	.810	.939	.334	.006
Error	131.980	153	.863			
Total	2734.000	161				
Corrected Total	141.975	160				

a. R Squared = .070 (Adjusted R Squared = .028)

(d) Satisfaction

Tests of Between-Subjects Effects

Dependent Variable: Qua_Sufficiency

Source	Type III Sum of Squares	df	Mean Square	F	Sig.	Partial Eta Squared
Corrected Model	19.871 ^a	7	2.839	3.299	.003	.131
Intercept	2283.693	1	2283.693	2654.161	<.001	.945
factual_counterfactual	10.525	1	10.525	12.233	<.001	.074
explanation_type	.263	1	.263	.305	.581	.002
with_text	8.410	1	8.410	9.775	.002	.060
factual_counterfactual * explanation_type	.033	1	.033	.038	.846	.000
factual_counterfactual * with_text	1.739	1	1.739	2.021	.157	.013
explanation_type * with_text	.698	1	.698	.812	.369	.005
factual_counterfactual * explanation_type * with_text	.192	1	.192	.224	.637	.001
Error	131.644	153	.860			
Total	2570.000	161				
Corrected Total	151.516	160				

a. R Squared = .131 (Adjusted R Squared = .091)

(e) Sufficiency

Tests of Between-Subjects Effects

Dependent Variable: Qua_Completeness

Source	Type III Sum of Squares	df	Mean Square	F	Sig.	Partial Eta Squared
Corrected Model	21.491 ^a	7	3.070	2.829	.008	.115
Intercept	2130.735	1	2130.735	1963.576	<.001	.928
factual_counterfactual	15.980	1	15.980	14.726	<.001	.088
explanation_type	.856	1	.856	.789	.376	.005
with_text	3.995	1	3.995	3.682	.057	.023
factual_counterfactual * explanation_type	.006	1	.006	.006	.941	.000
factual_counterfactual * with_text	.163	1	.163	.150	.699	.001
explanation_type * with_text	.272	1	.272	.251	.617	.002
factual_counterfactual * explanation_type * with_text	.495	1	.495	.456	.501	.003
Error	166.025	153	1.085			
Total	2431.000	161				
Corrected Total	187.516	160				

a. R Squared = .115 (Adjusted R Squared = .074)

(f) Completeness

Fig. B.14. Factorial ANOVA results for subjective quality metrics. The counterfactual nature of the explanation (FCCE vs. ICE) is a significant driver of subjective quality across most metrics.

For classifier guidance, we used the corresponding pretrained ImageNet classifier with image size 256×256 , attention resolutions at 32, 16, 8, classifier depth 2, classifier width 128, attention pooling, and residual up/down blocks enabled. Classifier guidance was applied at every denoising step with guidance scale set to 2.0. The classifier was executed in float32 precision for implementation compatibility.

For benchmarking, each source image was represented as a 256×256 tensor normalized to the $[-1, 1]$ range. It was then forward-noised to the starting point of the respaced diffusion trajectory, after which classifier-guided DDIM reverse sampling was performed toward the target class. The target class was not randomly sampled; instead, it was determined by the class-pair evaluation protocol associated with each source instance.

This benchmark was conducted only in the ImageNet setting, because the required pretrained diffusion model and corresponding classifier are directly available only in that setting, whereas comparable off-the-shelf source-conditioned diffusion counterfactual models are not readily available for CUB or MNIST. We therefore used the publicly available pretrained ImageNet model without retraining or dataset-specific adaptation, so that the reported runtime reflects the practical inference cost of source-conditioned diffusion-based counterfactual generation rather than additional training overhead.

For reproducibility, all measurements were taken after one warm-up run and then averaged over repeated executions on the same NVIDIA Tesla T4 GPU. The reported latency corresponds to average per-image generation time under this fixed inference protocol.

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